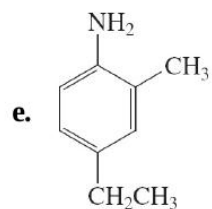
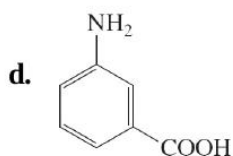
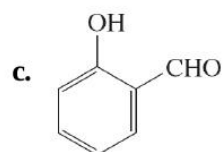
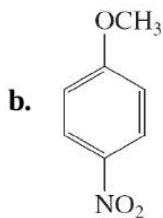
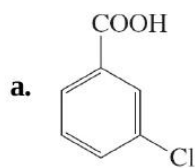
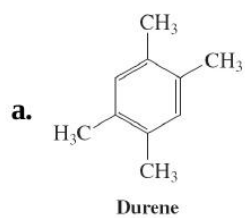


36. Name each of the following compounds by using the IUPAC system and, if possible, a reasonable common alternative. (**Hint:** The order of functional group precedence is $\text{-COOH} > \text{-CHO} > \text{-OH} > \text{-NH}_2$.
 $\text{-COOH} > \text{-CHO} > \text{-OH} > \text{-NH}_2$.)



37. Give a proper IUPAC name for each of the following commonly named substances.

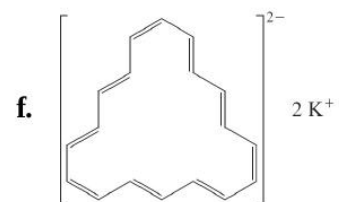
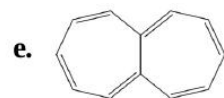
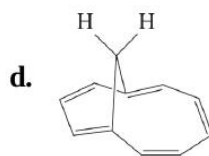
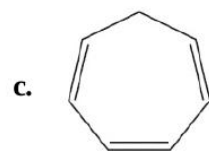
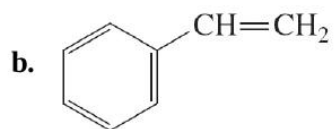


38. Draw the structure of each of the following compounds. If the name itself is incorrect, give a correct systematic alternative. **(a)** *o*-Chlorobenzaldehyde; **(b)** 2,4,6-trihydroxybenzene; **(c)** 4-nitro-*o*-xylene; **(d)** *m*-isopropylbenzoic acid; **(e)** 4,5-dibromoaniline; **(f)** *p*-methoxy-*m*-nitroacetophenone.

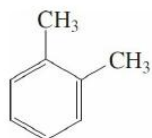
39. The complete combustion of benzene is exothermic by approximately $-789 \text{ kcal mol}^{-1}$. What would this value be if benzene lacked aromatic stabilization?

41. Complete hydrogenation of 1,3,5,7-cyclooctatetraene is exothermic by

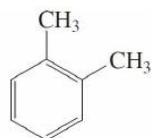
42. Which of the following structures qualify as being aromatic, according to Hückel's rule?



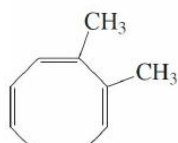
58. Because of cyclic delocalization, structures A and B shown here for *o*-dimethylbenzene (*o*-xylene) are simply two resonance forms of the same molecule. Can the same be said for the two dimethylcyclooctatetraene structures C and D? Explain.



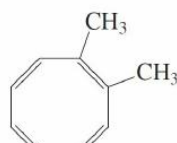
A



B

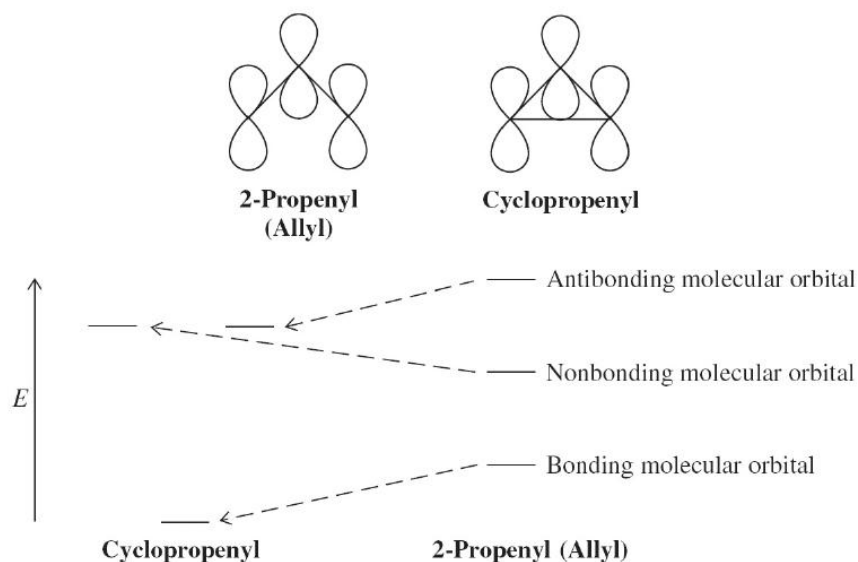


C

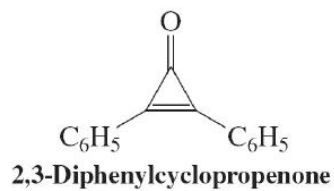


D

59. The energy levels of the 2-propenyl (allyl) and cyclopropenyl $\pi\pi$ systems are compared qualitatively in the diagram on the next page. **(a)** Draw the three molecular orbitals of each system, using plus and minus signs and dotted lines to indicate bonding overlap and nodes, as in [Figure 15-4](#). Does either of these systems possess degenerate molecular orbitals? **(b)** How many $\pi\pi$ electrons would give rise to the maximum stabilization of the cyclopropenyl system, relative to 2-propenyl (allyl)? (Compare [Figure 15-5](#), for benzene.) Draw Lewis structures for both systems with this number of π electrons and any appropriate atomic charges. **(c)** Could the cyclopropenyl system drawn in (b) qualify as being “aromatic”? Explain.

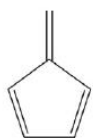


60. **CHALLENGE** 2,3-Diphenylcyclopropenone (see structure below) forms an addition product with HBr that exhibits the properties of an ionic substance (i.e., a salt). Suggest a structure for this product and a reason for its existence as a stable entity.

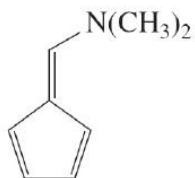


61. **CHALLENGE** Is cyclobutadiene dication ($\text{C}_4\text{H}_4^{2+}$) aromatic according to Hückel's rule? Sketch its π molecular orbital diagram to illustrate your answer.

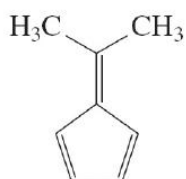
62. All the molecules shown below are examples of “fulvenes,” or methylenecyclopentadienes.



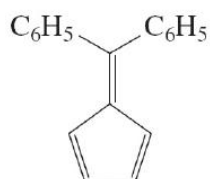
5-Methyldiene-
1,3-cyclopentadiene
“Fulvene”



6-Dimethylamino-
fulvene



6,6-Dimethyl-
fulvene



6,6-Diphenyl-
fulvene

- a. One of these structures is considerably more acidic than the others, with a $\text{p}K_{\text{a}}$ around 20. Identify it and its most acidic hydrogen(s), and explain why it is an unusually strong acid for a molecule with only carbon-hydrogen bonds.
- b. None of the 7-methyldiene-1,3,5-cycloheptatrienes that correspond to the fulvene structures above show any unusual acidity. Explain.

- 63.** A characteristic reaction of fulvenes is nucleophilic addition. To which carbon in a fulvene would you expect nucleophiles to add, and why?

64. **CHALLENGE** (a) The ^1H NMR spectrum of [18]annulene shows two signals, at $\delta = 9.28$ (12 H) and -2.99 (6 H) ppm. The negative chemical shift value refers to a resonance upfield (to the right) of $(\text{CH}_3)_4\text{Si}$. Explain this spectrum. (**Hint:** Consult [Figure 15-9](#).) (b) The unusual molecule 1,6-methano[10]annulene (shown below) exhibits two sets of signals in the ^1H NMR spectrum at $\delta = 7.10$ (8 H)

65. The ^1H NMR spectrum of the most stable isomer of [14]annulene shows two signals, at $\delta = -0.61$ (4 H) and 7.88 (10 H) ppm. Two possible structures for [14]annulene are shown here. How do they differ? Which one corresponds to the NMR spectrum described?

